① * Form a P.D.E by eliminating. The constants from (a) $(x-a)^2+(y-b)^2=z^2\cot^2\alpha$ where α is a parameter.

Sol:- Given eqn is $(x-a)^2+(y-b)^2=z^2\cot^2\alpha$ —— ①

Differentiating in eqn ① partially w.r.t x we get

$$2(x-a)+0=2z\frac{\partial z}{\partial x}\cot^2\alpha$$

$$x-a=zp\cot^2\alpha \quad \text{—— ②}$$

Differentiating in eqn ① partially w.r.t y we get

$$2(y-a)=2z\frac{\partial z}{\partial y}\cot^2\alpha$$

$$y-a=zq\cot^2\alpha \quad \text{—— ③}$$

Sub eqn ②, ③ in ① we get

$$① \Rightarrow (zp\cot^2\alpha)^2+(zq\cot^2\alpha)^2=z^2\cot^2\alpha$$

$$z^2p^2\cot^4\alpha+z^2q^2\cot^4\alpha=z^2\cot^2\alpha$$

$$(p^2+q^2)z^2\cot^4\alpha=z^2\cot^2\alpha$$

$$\boxed{(p^2+q^2)\cot^2\alpha=1}$$

Which is required. PDE

Solve $p\tan x + q\tan y = \tan z$

Given that $p\tan x + q\tan y = \tan z$

This is of the form $pP + qQ = R$

Here $P = \tan x$, $Q = \tan y$, $R = \tan z$

We know that, the subsitory eqn

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$$

$$\frac{dx}{\tan x} = \frac{dy}{\tan y} = \frac{dz}{\tan z}$$

Taking first two of the above eqn

$$\int \frac{dx}{\tan x} = \int \frac{dy}{\tan y} \Rightarrow \int \cot x\, dx = \int \cot y\, dy$$

$$\log|\sin x| = \log|\sin y| + \log c_1$$

$$c_1 = \left|\frac{\sin x}{\sin y}\right|$$

Taking last two of the above eqn

$$\int \frac{dy}{\tan y} = \int \frac{dz}{\tan z} \Rightarrow \int \cot y\, dy = \int \cot z\, dz$$

$$\log|\sin y| - \log|\sin z| = \log c_2 \Rightarrow c_2 = \frac{\sin y}{\sin z}$$

Hence the complete solution is

$$\phi\left(\frac{\sin x}{\sin y}, \frac{\sin y}{\sin z}\right) = 0 \quad (\text{or}) \quad \frac{\sin x}{\sin y} = \phi\left(\frac{\sin y}{\sin z}\right)$$

t octant.

* Form PDE by eliminating a,b,c from 5-1.2

(a) $x^2/a^2 + y^2/b^2 + z^2/c^2 = 1$

Sol:- Given that $x^2/a^2 + y^2/b^2 + z^2/c^2 = 1$ — ①

Diff eqn ① Partially w.r.t 'x'

$$\frac{2x}{a^2} + \frac{2z}{c^2}\frac{\partial z}{\partial x} = 0$$

$$\frac{x}{a^2} + \frac{z}{c^2}\frac{\partial z}{\partial x} = 0 \Rightarrow \frac{x^2}{a^2} + \frac{z}{c^2}P = 0 \quad \text{— ②}$$

Diff eqn ① Partially w.r.t 'y'

$$0 + \frac{2y}{b^2} + \frac{2z}{c^2}\frac{\partial z}{\partial y} = 0$$

$$\frac{y}{b^2} + \frac{z}{c^2}q = 0 \quad \text{— ③}$$

Since, it is not possible to eliminate a,b,c from the relation eqn ①, eqn ② and eqn ③

The required one more relation differentiate eqn ② partially w.r.t 'x' we get,

$$\frac{1}{a^2} + \frac{1}{c^2}\left[z\frac{\partial^2 z}{\partial x^2} + P\frac{\partial z}{\partial x}\right] = 0$$

$$\frac{1}{a^2} + \frac{1}{c^2}\left[zr + P\right] = 0 \quad \text{— ④}$$

Multiplying eqn ④ by 'x' and then substract eqn ② from it.

$$\frac{x}{a^2} + \frac{x}{c^2}zr + \frac{xP^2}{c^2} - \frac{x^2}{a^2} - \frac{zP}{c^2} = 0$$

$$xzr + xP^2 - zP = 0$$

$$\boxed{Pz = xP^2 + xzr}$$

(b) Which is the required PDE

2(b) Which is the required PDE

* Form the PDE by eliminating a and b from $\log(az-1) = x+ay+b$.

Sol: Given that $\log(az-1) = x+ay+b$ — ①

Differentiate eqn ① Partially w.r.t 'x' we get

$$\frac{1}{az-1}\, a\frac{\partial z}{\partial x} = 1 \Rightarrow ap = az-1 \quad \text{— ②}$$

Diff eqn ① Partially w.r.t 'y', we get

$$\frac{1}{az-1} \cdot a\frac{\partial z}{\partial y} = a$$

$$q = az - 1$$

$$q = ap \Rightarrow a = q/p \quad (\text{eqn } ②)$$

Sub $a = q/p$ in eq ②

$$② \Rightarrow \frac{q}{p} \times p = \frac{q}{p} \times z - 1$$

$$qp = qz - p$$

$$qz = qp + p$$

$$\boxed{qz = p(q+1)}$$

Which is the required PDE

3/ * Solve $(x^2-yz)P+(y^2-zx)Q=z^2-xy$.

Given that $(x^2-yz)P+(y^2-zx)Q=z^2-xy$

This is in the form of $Pp+Qq=R$

Here $P=x^2-yz$, $Q=y^2-zx$, $R=z^2-xy$

Here the subsidary equation

$$\frac{dx}{P}=\frac{dy}{Q}=\frac{dz}{R}=\frac{l\,dx+m\,dy+n\,dz}{lP+mQ+nR}$$

Using the multipliers are 1,-1,0 and 0,1,-1, as multipliers we have.

Each fraction $\cancel{\frac{1}{x}(x^2-yz)+\frac{1}{y}(y^2-zx)+\frac{1}{z}(z^2-xy)}$

$$\frac{dx-dy}{(x^2-yz)-(y^2-zx)}=\frac{dy-dz}{(y^2-zx)-(z^2-xy)}$$

$$\frac{dx-dy}{(x^2-y^2)+z(x-y)}=\frac{dy-dz}{(y^2-z^2)+x(y-z)}$$

$$\frac{d(x-y)}{(x-y)(x+y)+z(x-y)}=\frac{d(y-z)}{(y+z)(y-z)+x(y-z)}$$

$$\frac{d(x-y)}{(x-y)}=\frac{d(y-z)}{(y-z)}$$

Intigrating on both sides

$$\log(x-y)=\log(y-z)+\log c_1$$

$$c_1=\frac{x-y}{y-z}$$

Now again taking x, y, z and $1, 1, 1$ as multipliers we have

Each fraction $= \dfrac{x\,dx + y\,dy + z\,dz}{x(x^2-yz)+y(y^2-zx)+z(z^2-xy)} = \dfrac{dx+dy+dz}{x^2-yz+y^2-zx+z^2-xy}$

$$\frac{x\,dx+y\,dy+z\,dz}{x^3+y^3+z^3-3xyz} = \frac{dx+dy+dz}{x^2+y^2+z^2-xy-yz-zx}$$

$$\frac{x\,dx+y\,dy+z\,dz}{(x+y+z)(x^2+y^2+z^2-xy+yz-zx)} = \frac{dx+dy+dz}{x^2+y^2+z^2-xy-yz-zx}$$

$$\frac{x\,dx+y\,dy+z\,dz}{x+y+z} = dx+dy+dz$$

$$x\,dx+y\,dy+z\,dz = (x+y+z)(dx+dy+dz)$$

Integrating on both sides

$$\int x\,dx + \int y\,dy + \int z\,dz = \int (x+y+z)(dx+dy+dz)$$

$$\frac{x^2}{2} + \frac{y^2}{2} + \frac{z^2}{2} = \frac{(x+y+z)^2}{2} + C_2$$

$$C_2 = \frac{(x^2+y^2+z^2) - (x^2+y^2+z^2+2xy+2yz+2zx)}{2}$$

$$C_2 = xy+yz+zx.$$

Hence the complete solution

$$\phi(C_1, C_2) = 0$$

$$\phi\left(\frac{x-y}{y-z},\ xy+yz+zx\right) = 0$$

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* Solve $(mz-ny)p+(nx-lz)q=(ly-mx)$.

Given that $(mz-ny)P+(nx-lz)q=(ly-mx)$

This is of the form $pP+qQ=R$

Here $P=mz-ny$, $Q=(nx-lz)$, $R=ly-mx$.

Here the subsidary eqn

$$\frac{dx}{P}=\frac{dy}{Q}=\frac{dz}{R}=\frac{ldx+mdy+ndz}{}$$

using the multipliers are x, y, z

$$\text{Each fraction}=\frac{xdx+ydy+zdz}{x(mz-ny)+y(nx-lz)+z(ly-mx)}$$

$$=\frac{xdx+ydy+zdz}{mxz-xny+ynx-ylz+zly-zmx}$$

$$=\frac{xdx+ydy+zdz}{0}$$

$$xdx+ydy+zdz=0$$

Integrating on both sides

$$\int xdx+\int ydy+\int zdz=0$$

$$\frac{x^2}{2}+\frac{y^2}{2}+\frac{z^2}{2}=c_1$$

We choose other multipliers are l, m, n

$$\text{Each fraction}=\frac{ldx+mdy+ndz}{l(mz-ny)+m(nx-lz)+n(ly-mx)}$$

$$=\frac{ldx+mdy+ndz}{\cancel{lmz}-\cancel{lny}+\cancel{mnx}-\cancel{mlz}+\cancel{nly}-\cancel{mnx}}$$

$$=\frac{ldx+mdy+ndz}{0}$$

5(a) Form the PDE by eliminating the arbitrary function from $Z = yf(x) + x\,g(y)$

Sol:- Given $Z = yf(x) + x\,g(y)$ —— (1)

Diff eqn (1) p.w.r.to 'x' we get

$$\frac{\partial z}{\partial x} = yf'(x) + g(y) \quad \text{—— (2)}$$

Diff eqn (2) p.w.r.to x we get

$$\frac{\partial^2 z}{\partial x^2} = yf''(x)$$

Diff eqn (1) w.r.to 'y' we get

$$\frac{\partial z}{\partial y} = f(x) + x\,g'(y) \quad \text{—— (4)}$$

Diff eqn (4) w.r.to 'x' we get.

$$\frac{\partial^2 z}{\partial x \partial y} = f'(x) + g'(y) \quad \text{—— (5)}$$

Diff eqn (4) p w.r.to 'y' we get

$$\frac{\partial^2 z}{\partial y^2} = x\,g''(y) \quad \text{—— (6)}$$

from (2);

$$f'(x) = \frac{1}{y}\left[p - g(y)\right] \quad \text{—— (7)}$$

from eqn (4)

$$g'(y) = \frac{1}{x}\left[q - f(x)\right] \quad \text{—— (8)}$$

Sub eqn (7), (8) in eqn (5)

$$(5) \Rightarrow \frac{\partial^2 z}{\partial x \partial y} = \frac{p}{y} - \frac{g(y)}{y} + \frac{q}{x} - \frac{f(x)}{x}$$

$$xy\,s = px - g(y)\cdot x + qy - y f(x)$$

$$sxy = px + qy - [x\,g(y) + y f(x)]$$

$$sxy = px + qy - z \quad (\because \text{eqn } (1))$$

Which is required PDE

5(b) Solve the PDE $p\sqrt{x} + q\sqrt{y} = \sqrt{z}$

Sol:- Given that $p\sqrt{x} + q\sqrt{y} = \sqrt{z}$

This is of that form $pP + qQ = R$

Here $P = \sqrt{x}$, $Q = \sqrt{y}$, $R = \sqrt{z}$

We know that the subsitory eqn

$$\frac{dx}{\sqrt{x}} = \frac{dy}{\sqrt{y}} = \frac{dz}{\sqrt{z}}$$

Taking first two of the above eqn

$$\int \frac{dx}{\sqrt{x}} = \int \frac{dy}{\sqrt{y}}$$

$$2\sqrt{x} = 2\sqrt{y} + 2\sqrt{c_1}$$

$$\sqrt{x} = \sqrt{y} + \sqrt{c_1} \Rightarrow c_1 = (\sqrt{x} - \sqrt{y})$$

Taking last two of the above eqn

$$\int \frac{dy}{\sqrt{y}} = \int \frac{dz}{\sqrt{z}}$$

$$2\sqrt{y} = 2\sqrt{z} + 2c_2$$

$$c_2 = \sqrt{y} - \sqrt{z}$$

Hence the complete solution is

$$\phi(\sqrt{x} - \sqrt{y}, \sqrt{y} - \sqrt{z}) = 0 \quad \text{or}$$

$$\sqrt{x} - \sqrt{y} = \phi(\sqrt{y} - \sqrt{z})$$

6* Solve $(y+z)p - (z+x)q = x-y$

Sol:- Given that $(y+z)p - (z+x)q = x-y$

This is in the form of $Pp + Qq = R$

$$P = y+z \qquad Q = -z-x \qquad R = x-y$$

Here the subsidary equation

$$\frac{dx}{y+z} = \frac{dy}{-z-x} = \frac{dz}{x-y} = \frac{l\,dx + m\,dy + n\,dz}{lP + mQ + nR}$$

. Using the multipliers are 1, 1, 1

$$\text{Each fraction} = \frac{dx+dy+dz}{y+z-z-x+x-y}$$

$$= \frac{dx+dy+dz}{0}$$

$$dx+dy+dz = 0$$

$$\int dx + \int dy + \int dz = c_1$$

$$c_1 = x+y+z$$

We choose other multipliers are x, y, z

$$\text{Each fraction} = \frac{x\,dx + y\,dy - z\,dz}{x(y+z) - y(z+x) - z(x-y)}$$

$$= \frac{x\,dy + y\,dz - z\,dz}{xy + xz - yz - xy - xz + yz}$$

$$= \frac{x\,dy + y\,dz - z\,dz}{0}$$

$$\int x\,dx + \int y\,dy - \int z\,dz = 0$$

$$\frac{x^2}{2} + \frac{y^2}{2} - \frac{z^2}{2} = c_2$$

$$c_2 = x^2 + y^2 - z^2$$

Hence the general solution

$$\phi(c_1, c_2) = 0$$

$$\phi(x+y+z,\ x^2+y^2-z^2) = 0$$

Find the general solution of

which

7(a) Form the PDE by eleminating the arbitrary Function from $xyz = f(x^2+y^2+z^2)$

Sol:- Given that $xyz = f(x^2+y^2+z^2)$ — (1)

Differentiate eqn (1) P w.r.t 'x' we get

$$y\left[x\frac{\partial z}{\partial x}+z\cdot 1\right]=f'(x^2+y^2+z^2)\left(2x+2z\frac{\partial z}{\partial x}\right)$$

$$y(xp+z)=f'(x^2+y^2+z^2)\,2(x+zp)$$

$$y(xp+z)=2(x+zp)\,f'(x^2+y^2+z^2) \quad \text{--- (2)}$$

Diff eqn (1) p.w.r.t 'y' we get

$$x\left[y\frac{dz}{dy}+z\cdot 1\right]=f'(x^2+y^2+z^2)\left(2y+2z\frac{\partial z}{\partial y}\right)$$

$$x(yq+z)=2(y+zq)\,f'(x^2+y^2+z^2) \quad \text{--- (3)}$$

$$\frac{(2)}{(3)} \quad \frac{y(xp+z)}{x(yq+z)}=\frac{2(x+zp)\,f'(x^2+y^2+z^2)}{2(y+zq)\,f'(x^2+y^2+z^2)}$$

$$(xyp+yz)(y+zq)=(x+zp)(xyq+xz)$$

$$xy^2p+xyzpq+y^2z+yqz^2-x^2yq-zxypq-x^2z-xpz^2=0$$

$$(-x^2+y^2)z+(yp-xp)z^2+xy(yp-xq)=0$$

Which is the required PDE.

7(b) Form the PDE by eleminating the arbitrary functions from $z = f(2x+y) + g(3x-y)$.

Given that $z = f(2x+y) + g(3x-y)$ — ①

Diff eqn ① P.w.r.to 'x' we get

$$\frac{\partial z}{\partial x} = 2f'(2x+y) + 3g'(3x-y) \quad \text{— ②}$$

Diff eqn ① P.w.r.to 'y' we get

$$\frac{\partial z}{\partial y} = f'(2x+y) - g'(3x-y) \quad \text{— ③}$$

Diff eqn ② P.D.w.r.to 'x' we get

$$\frac{\partial^2 z}{\partial x^2} = 2\times 2f''(2x+y) + 3\times 3g''(3x-y) -$$

$$\frac{\partial^2 z}{\partial x^2} = 4f''(2x+y) + 9g''(3x-y) \quad \text{—④}$$

Diff eqn ③ P.D.w.r.to 'y' we get

$$\frac{\partial^2 z}{\partial y^2} = f''(2x+y) + g''(3x-y) \quad \text{—⑤}$$

Diff eqn ③ P.w.r.to 'x' we get

$$\frac{\partial^2 z}{\partial x \partial y} = 2f''(2x+y) - 3g''(3x-y) \quad \text{—⑥}$$

③+⑥ $\frac{\partial^2 z}{\partial x \partial y} + \frac{\partial^2 z}{\partial y} = f''(2x+y) + g''(3x-y) + 2f''(2x+y) - 3g''(3x-y)$

$$s + r = 3f''(2x+y) - 2g''(3x-y)$$

Q. Solve $\frac{\partial^2 z}{\partial x^2} - \frac{\partial^2 z}{\partial x \partial y} = \cos x \cos 2y$

$\cos A \cos B = \frac{1}{2}(\cos(A+B) + \cos(A-B))$

Sol:- Given that $\frac{\partial^2 z}{\partial x^2} - \frac{\partial^2 z}{\partial x \partial y} = \cos x \cos 2y$

Can be written in $(D^2 - DD')z = \cos x \cos 2y$

This can be written in $f(D, D')z = \cos(ax+by)$

$$f(D, D') = D^2 - DD'$$

$$f(m, 1) = 0$$

$$m^2 - m = 0$$

$$m(m-1) = 0$$

$$m = 0, 1$$

$$Z_c = \phi_1(y) + \phi_2(y+x)$$

$$Z_p = \frac{1}{2(D^2 - DD')} 2\cos x \cos 2y$$

$$= \frac{1}{2(D^2 - DD')}(\cos(x+2y) + \cos(x-2y))$$

$$= \frac{1}{2}\left[\frac{1}{D^2 - DD'}\cos(x+2y) + \frac{1}{-DD' + D^2}\cos(x-2y)\right]$$

$$= \frac{1}{2}\left[\frac{1}{-1+2}\cos(x+2y) + \frac{1}{-1-2}\cos(x-2y)\right]$$

$$= \frac{1}{2}\left[\frac{1}{1}\cos(x+2y) - \frac{1}{3}\cos(x-2y)\right]$$

$$= \frac{1}{2}\cos(x+2y) - \frac{1}{6}\cos(x-2y)$$

Hence the general solution is

$$Z = Z_c + Z_p$$

$$= \phi_1(y) + \phi_2(y+x) + \frac{1}{2}\cos(x+2y) - \frac{1}{6}\cos(x-2y).$$

9

* Solve $(x^2-y^2-z^2)p+(2xy)q=2xz$

Sol:- Given that $(x^2-y^2-z^2)p+2xyq=2xz$

This is in the form of $Pp+Qq=R$.

Here $P=x^2-y^2-z^2$, $Q=2xy$, $R=2xz$

Here the subsidiary equations

$$\frac{dx}{P}=\frac{dy}{Q}=\frac{dz}{R}=\frac{l\,dx+m\,dy+n\,dz}{lP+mQ+nR}$$

Taking fractions 2, 3 in above equation

$$\frac{dy}{2xy} = \frac{dz}{2zx}$$

$$\frac{dy}{y} = \frac{dz}{z}$$

$$\int \frac{dy}{y} = \int \frac{dz}{z}$$

$$\log y = \log z + \log c_1$$

$$\log y - \log z = \log c_1$$

$$\log\left(\frac{y}{z}\right) = \log c_1$$

$$\therefore c_1 = y/z$$

Using the multipliers are x, y, z ~~are~~ and also taking last two in the above eqn we have

$$\frac{dz}{2xz} = \frac{l\,dx + m\,dy + n\,dz}{lP + mQ + nR}$$

$$\frac{dz}{2xz} = \frac{x\,dx + y\,dy + z\,dz}{x(x^2-y^2-z^2) + y(2xy) + z(2xz)}$$

$$\frac{dz}{2xz} = \frac{x\,dx + y\,dy + z\,dz}{x^3 - xy^2 - xz^2 + 2xy^2 + 2xz^2}$$

$$\frac{dz}{2xz} = \frac{x\,dx + y\,dy + z\,dz}{x^3 + xy^2 + xz^2}$$

$$\frac{dz}{2xz} = \frac{x\,dx + y\,dy + z\,dz}{x(x^2+y^2+z^2)}$$

$$\frac{dz}{2z} = \frac{x\,dx + y\,dy + z\,dz}{x^2+y^2+z^2}$$

$$\frac{dz}{z} = \frac{2x\,dx + 2y\,dy + 2z\,dz}{x^2+y^2+z^2}$$

integrating on both sides

$$\int \frac{dz}{z} = \int \frac{2x\,dx + 2y\,dy + 2z\,dz}{x^2+y^2+z^2}$$

$$\log z + \log c_2 = \log(x^2+y^2+z^2)$$

$$\log c_2 = \log\left(\frac{x^2+y^2+z^2}{z}\right)$$

$$c_2 = (x^2+y^2+z^2)/z$$

Hence the complete solution

$$\phi(c_1, c_2) = 0$$

$$\phi\left(y/z, \frac{x^2+y^2+z^2}{z}\right) = 0$$

10

Q. Find the differential equation of all spheres of radius 8 and having their centers in the yz plane.

Sol:- Any point in yz plane is of the form $(0, a, b)$.

The eqn of the sphere with centre $(0, a, b)$ and

radius 8 is given by

$$x^2+(y-a)^2+(z-b)^2=64 \quad \text{—} (1)$$

diff. in eqn (1) Partially w.r.t x & y. We get

$$(1) \Rightarrow 2x+2(z-b)\frac{\partial z}{\partial x}=0$$

$$x+(z-b)P=0 \qquad \left(\because \frac{\partial z}{\partial x}=P\right)$$

$$z-b=\frac{-x}{P} \quad \text{—} (2)$$

$$(1) \Rightarrow 2(y-a)+2(z-b)\frac{\partial z}{\partial y}=0$$

$$(y-a)+(z-b)q=0$$

$$(y-a)+\left(\frac{-x}{P}\right)q=0$$

$$y-a=\frac{xq}{P} \quad \text{—} (3)$$

Sub eqn (2) & (3) in eqn (1)

$$(1) \Rightarrow x^2+\frac{x^2q^2}{P^2}+\frac{x^2}{P^2}=64$$

$$P^2x^2+q^2x^2+x^2=64P^2$$

$$\boxed{x^2(P^2+q^2+1)=64P^2}$$

Find the general solution of the partial differential equation. $y(x-z)p+(z^2-xz-x^2)q = y(2x-z)$. Hence obtained the particular solution which passes through the ellipse $z=0$, $2x^2+4y^2=$

Given that $y(x-z)p+(z^2-xz-x^2)q = y(2x-z)$ —— ①

This is in the form of $pP+qQ=R$

Here $P = yx-zy$, $Q = z^2-xz-x^2$, $R = 2xy-zy$

The subsidary equation is given by

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} = \frac{ldx+mdy+ndz}{lP+mQ+nR}$$

$$\frac{dx}{xy-yz} = \frac{dy}{z^2-xz-x^2} = \frac{dz}{2xy-yz} = \frac{ldx+mdy+ndz}{lP+mQ+nR}$$

Taking x, y, z as multipliers we have;

Each fraction $= \dfrac{xdx+ydy+zdz}{xy(x-z)+y(z^2-xz-x^2)+zy(2x-z)}$

$$= \frac{xdx+ydy+zdz}{x^2y-xyz+yz^2-xzy-yx^2+2xyz-z^2y}$$

$$= \frac{xdx+ydy+zdz}{0}$$

$$xdx+ydy+zdz = 0$$

$$\int xdx+\int ydy+\int zdz = 0$$

$$\frac{x^2}{2}+\frac{y^2}{2}+\frac{z^2}{2} = c_1$$

$$c_1 = \frac{x^2+y^2+z^2}{2} \quad \text{—— ②}$$

Consider first & third fractions on the above eqn

$$\frac{dx}{y(x-z)} = \frac{dz}{y(2x-z)}$$

$$\frac{dx}{x-z} = \frac{dz}{2x-z}$$

$$(2x-z)dx = dz(x-z)$$

$$2xdx - zdx - xdz + zdz = 0$$

$$2xdx + zdz - (xdz + zdx) = 0$$

$$2xdx + zdz - d(xz) = 0$$

Integrating on both sides

$$\int 2xdx + \int zdz - \int d(xz) = 0$$

$$x^2 + \frac{z^2}{2} - xz = C_2 \quad \text{—} (3)$$

General solution

$$\phi(C_1, C_2) = 0$$

$$\phi\left(\frac{x^2}{2} + \frac{y^2}{2} + \frac{z^2}{2}, x^2 + \frac{y^2}{2} - xz\right) = 0$$

also the given that the surface passes through the ellipse $z=0$, $2x^2 + y^2 = 1$ — (4)

from (2) ; $x^2 + y^2 = C_1$ — (5)

from (3) ; $x^2 = C_2$ — (6)

eqn (5)+(6) $\Rightarrow 2x^2 + y^2 = C_1 + C_2$ — (7)

from eq (4) and (7)

$$C_1 + C_2 = 1 - 3y^2 \quad \text{—} (8)$$

substitute eqn (2), (3) in (8)

(8) $\Rightarrow x^2 + y^2 + z^2 + x^2 + \frac{z^2}{2} - xz = 1 - 3y^2$

$$2x^2 + 4y^2 + \frac{3z^2}{2} - xz = 1$$

2. Form the PDE by eleminating the arbitrary functions ϕ from $\phi(y/x, x^2+y^2+z^2)=0$

Sol:- Given that $\phi(y/x, x^2+y^2+z^2)=0$

This is of the form $\phi(u,v)=0$

Here $u = y/x$ $\quad$ $v = x^2+y^2+z^2$

$$\frac{\partial u}{\partial x} = -y/x^2 \qquad \frac{\partial v}{\partial x} = 2x$$

$$\frac{\partial u}{\partial y} = 1/x \qquad \frac{\partial v}{\partial y} = 2y$$

$$\frac{\partial u}{\partial z} = 0 \qquad \frac{\partial v}{\partial z} = 2z$$

We know that

$$\begin{vmatrix} \frac{\partial u}{\partial x} + p\frac{\partial u}{\partial z} & \frac{\partial v}{\partial x} + p\frac{\partial u}{\partial z} \\ \frac{\partial u}{\partial y} + q\frac{\partial u}{\partial z} & \frac{\partial v}{\partial y} + q\frac{\partial u}{\partial z} \end{vmatrix} = 0$$

$$\begin{vmatrix} -y/x^2 & 2x+2pz \\ 1/x & 2y+2pz \end{vmatrix} = 0$$

$$\frac{-y}{x^2}(2y+2pz) - \frac{1}{x}(2x+2pz) = 0$$

$$-y(2y+2pz) - x(2x+2pz) = 0$$

$$-2y^2 - 2pyz - 2x^2 - 2xpz = 0$$

$$x^2+y^2+yqz+xpz = 0$$

$$(xz)p + (yz)q + (y^2+x^2) = 0$$

Which is the required PDE